

# ALEX CHEN

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## EDUCATION

University of Southern California

Los Angeles, CA

Bachelor of Science in Computer Engineering and Computer Science | GPA: 3.93

May 2027

## TECHNICAL SKILLS

**Design Tools:** Cadence OrCAD Capture CIS, Virtuoso, Allegro, HSPICE, Spectre, Synopsys Laker, Altium, Vivado  
**Languages:** Verilog, C++, Python, MATLAB | **Core Competencies:** RTL design, signal integrity, custom layout, FPGA, timing closure

## EXPERIENCE

HQ Telecom

Shanghai, China

Hardware Engineer Intern

May 2026 – August 2026

- Designed schematics of HP's Omni Ultra laptop on Cadence Capture CIS, achieving low-power Modern Standby mode at 160 mW
- Contributed to 50+ EE design issues by reworking up to 0201 components and retesting on oscilloscopes and Ellisis protocol analyzers
- Interfaced with layout team to review and revise output brd files against Intel PDG guides, realizing a 10-layer PCB featuring USB4/Thunderbolt, LPDDR5X, Intel NovaLake CPU, retimers, and codecs with production-level SNR and crosstalk ratios

Originsilicon

Nanning, China

R&D Intern

June 2024 – September 2024

- Implemented 8+ custom analog layout blocks in Synopsys Laker for mobile LCD display driver ICs through tape-out
- Designed level shifters, output buffers, and switch arrays using common-centroid matching, guard rings, and dummy insertion
- Cut channel block height up to 50% through floorplan restructuring, achieving clean DRC/LVS checks on basic unit blocks

## PROJECTS

Multiply-and-Accumulate (MAC) Accelerator

Los Angeles, CA

Circuit Design & Arithmetic Unit Development

April 2025 – May 2025

- Engineered 16-bit MAC datapath in 45nm CMOS, meeting timing closure at 0.33 GHz with radix-4 Booth encoding
- Designed custom standard-cell library and verified functionality with Cadence Virtuoso, Spectre simulation, and DRC/LVS signoff
- Optimized 6-to-2 compressor tree for delay, power, and area, reducing critical-path delay 20%

Blackjack Game on Artix-7 FPGA

Los Angeles, CA

Digital Design & RTL Implementation

January 2025 – April 2025

- Designed Verilog RTL game logic on Artix-7 FPGA with multi-state FSM and LFSR-based card randomization
- Implemented button debouncing, seven-segment and VGA display drivers as modular synthesizable blocks
- Verified functionality through Verilog testbench simulation and closed timing under Vivado synthesis constraints

PawStation Smart Pet Feeder

Los Angeles, CA

Embedded Systems & Electrical Design

February 2026 – July 2026

- Built smart pet feeder using RPi, TMC2209-driven NEMA 17 stepper, and HX711 load-cell feedback for closed-loop dispensing
- Designed full power and interface stack from wall input to motor drivers, I<sup>2</sup>C/SPI peripherals, and DS3231 RTC scheduling
- Integrated REST API and MJPEG camera stream for mobile status tracking, feed scheduling, and history data analysis

In-Car Heads-Up Display System

Los Angeles, CA

Embedded Systems & Automotive Data Integration

November 2025 – February 2026

- Built Raspberry Pi-based in-car display overlaying Google Maps with real-time vehicle telemetry from OBD-II ports
- Implemented Python pipeline for ECU data parsing, synchronization, and low-latency updates on HDMI touchscreen
- Optimized the interface for glance-based driving with responsive rendering and clear real-time data display

Hardware Validation Dashboard

Shanghai, China

Full-Stack Application Development

May 2026 – August 2026

- Engineered full-stack Python app (REST API, SQLite, multithreaded server) tracking 1k+ hardware issues with 15 weekly users
- Built serverless ETL pipeline with custom ZIP/XML deserialization, SHA-1 fingerprinting, and multi-user sync
- Shipped native Windows client with single-instance locking (Win32 API, ctypes) and real-time ECharts analytics dashboards

## RESEARCH EXPERIENCE

Shofflab

Cleveland, OH

Research Assistant

February 2023 – September 2024

- Co-authored peer-reviewed *Biomaterials* publication on reducing intracortical microelectrode inflammation, cited 16+ times
- Measured implanted microelectrode arrays via electrochemical impedance spectroscopy (EIS), analyzed degradation in MATLAB
- Automated data-processing pipeline in Python/Pandas, cutting analysis time two hours per experiment across 80GB of datasets

## CERTIFICATES & ACTIVITIES

Google Data Analytics

May 2024

Member, ACM @ USC (Association for Computing Machinery)

March 2025 – Present

Member, Taiwanese American Organization

September 2024 – Present